Week-3 PSOSM 2023

1. Consider the below response:
   "To this segment, consumer privacy is very important; they feel that they have been victims of privacy invasions, they are pessimistic about the future of privacy protection, and about a third of them favour creating a general federal regulatory agency on consumer privacy."

   To which of the three Westin Privacy categories is "they" in the above sentence most likely to fall?
   a. Fundamentalists
   b. Pragmatists
   c. Unconcerned
   d. None of the above

2. Which of the following features is used by the credibility model- TweetCred?
   a. Tweet links
   b. Tweet Author
   c. Tweet network
   d. Tweet meta data

3. Cronbach's alpha
   a. determines the statistical significance of a research study.
   b. assess the reliability and internal consistency of a questionnaire.
   c. analyze the effect size of an intervention.
   d. calculate the standard deviation of a sample population.

4. Suppose we have five tweets (TW1, TW2, TW3, TW4, and TW5) with their relevances based on user feedback or relevance labels:

   TW1: Highly relevant tweet (Relevance = 2)
   TW2: Moderately relevant tweet (Relevance = 1)
   TW3: Not relevant tweet (Relevance = 0)
   TW4: Relevant tweet (Relevance = 1)
   TW5: Highly relevant tweet (Relevance = 2)

   Now, let's assume the ranking algorithm generates a ranked list of tweets:

   Rank 1: TW5 (Highly relevant tweet)
   Rank 2: TW1 (Highly relevant tweet)
   Rank 3: TW3 (Not relevant tweet)
   Rank 4: TW2 (Moderately relevant tweet)
   Rank 5: TW4 (Relevant tweet)

What would be NDCG@5 for this ranked list of tweets?

a. 0.5
b. .513
c. .975
d. None of the above

Solution:
Step 1: Calculate DCG@5 (Discounted Cumulative Gain at rank 5).
Step 2: Calculate IDCG@5 (Ideal Discounted Cumulative Gain at rank 5).
Step 3: Divide DCG@5 by IDCG@5 to get NDCG@5.

Step 1: Calculate DCG@5 DCG@5 = rel1 + rel2/log2(2) + rel3/log2(3) + rel4/log2(4) + rel5/log2(5)

Given relevances: rel1 = 2 (for TW5) rel2 = 2 (for TW1) rel3 = 0 (for TW3) rel4 = 1 (for TW2) rel5 = 1 (for TW4)

DCG@5 = 2 + 2/log2(2) + 0/log2(3) + 1/log2(4) + 1/log2(5) DCG@5 = 2 + 2/1 + 0 + 1/2 + 1/2
DCG@5 = 4 + 0.5 + 0.5 DCG@5 = 5

Step 2: Calculate IDCG@5 (Ideal DCG@5) To calculate IDCG@5, we consider the ideal ranking based on relevance:

Ideal ranking: TW5 (Highly relevant tweet) > TW1 (Highly relevant tweet) > TW2 (Moderately relevant tweet) > TW4 (Relevant tweet) > TW3 (Not relevant tweet)

IDCG@5 = rel1 + rel2/log2(2) + rel3/log2(3) + rel4/log2(4) + rel5/log2(5)
For ideal ranking: rel1 = 2 (for TW5) rel2 = 2 (for TW1) rel3 = 1 (for TW2) rel4 = 1 (for TW4) rel5 = 0 (for TW3)
IDCG@5 = 2 + 2/log2(2) + 1/log2(3) + 1/log2(4) + 0/log2(5) IDCG@5 = 2 + 2/1 + 1/1.585 + 1/2 + 0 IDCG@5 = 2 + 2 + 0.63 + 0.5 IDCG@5 = 5.13
Step 3: Calculate NDCG@5 NDCG@5 = DCG@5 / IDCG@5
NDCG@5 = 5 / 5.13
NDCG@5 ≈ 0.975 (rounded to three decimal places)
So, the NDCG@5 for the ranked list of tweets is approximately 0.975.

Read through the report “Privacy in India: Attitudes and Awareness V 2.0” at https://cdn.iiit.ac.in/cdn/precog.iiit.ac.in/research/privacyindia/PI_2012_Complete_Report.pdf And answer the following questions [5-7].

5. How many participants have changed their default settings on their social network?
   a. 75%
   b. 60%
   c. 50%
   d. 25%

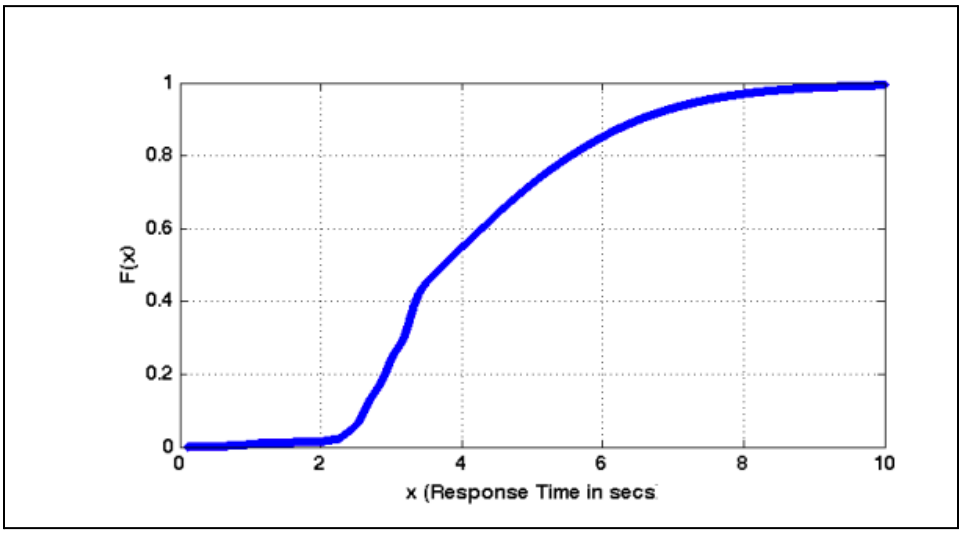
6. What is the primary purpose of a Focus Group Discussion (FGD) in the above report?

   a. To conduct one-on-one interviews with participants.
   b. To collect quantitative data through surveys and questionnaires.
   c. To observe participants' behaviours in a controlled environment.
   d. To facilitate group discussions and gather qualitative insights.

7. “When you hear the word privacy, what comes to mind”. Which is the most common respondent answer according to the above report?

   a. Information Privacy
   b. Bodily privacy
   c. Communication privacy
   d. Territorial privacy

8. Consider the below graph and choose the correct option:



a. For 82% of the users, the response time was less than 2 seconds, and for 99%, the response time was under 10 seconds.
b. For 99% of the users, the response time was less than 6 seconds, and for 100%, the response time was under 10 seconds.
c. For 82% of the users, the response time was less than 6 seconds, and for 99%, the response time was under 10 seconds.
d. None of the above

9. What is the Likert scale?

a. rating scale in surveys and questionnaires
b. gives the score for any real-time tweets
c. Interannotator agreement
d. None of the above

10. TweetCred is

a. a real-time web-based system to automatically evaluate the credibility of content on Twitter.
b. a real-time app-based system to automatically evaluate the score of tweets on Twitter.
c. a real-time app-based system to automatically assess the quality of content on Twitter.
d. None of the above